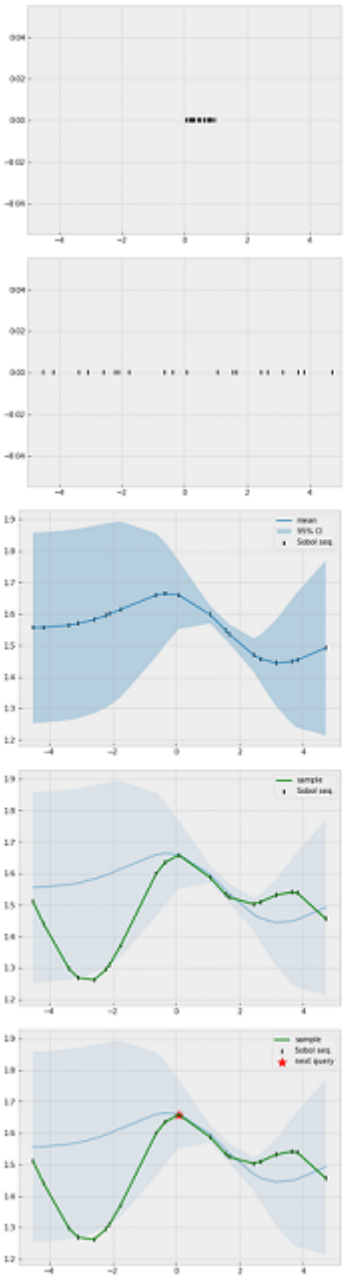


Figure 5.13 Flowchart of the implementation of Thompson sampling in BoTorch. We use a Sobol sequence to populate the search space, draw a sample from the Gaussian process on the sequence, and choose the point in the sequence that maximizes the sample to evaluate the objective function at.

Let's first discuss why we need to use a Sobol sequence to generate points that span our search space. A simpler solution is to use a dense grid. However, generating a dense grid quickly becomes intractable as the number of dimensions of our search space grows, so this strategy is out of the question. Another potential solution is to uniformly sample from that space, but statistical theory shows that uniform sampling is in fact not the best way to generate points that cover a space evenly, and a Sobol sequence does a better job.

Figure 5.14 shows the comparison between a Sobol sequence of 100 points and the same number of points uniformly sampled within the two-dimensional unit square. We see that The Sobol

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